

Qualification Pack



THREE-D Printing Operator

QP Code: MES/Q0511

Version: 2.0

NSQF Level: 4

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Qualification Pack

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MES/Q0511: THREE-D Printing Operator

Brief Job Description

A 3D printing operator is a competent individual with technical and creative skills in additive manufacturing. They use computer-aided (CAD) software to develop designs to be 3D printed into finished works. 3D Printing technicians assist in the designing and programming of products, ranging from prosthetic products to 3D miniatures. They may also provide 3D printing maintenance, check 3D renders for customers and run 3D printing tests. 3D printing technicians can also repair, maintain and clean 3D printers.

Personal Attributes

Individual at this job role have creative and knowledge to draw free hand, being great a 3D modeling tools, managing 3D printers. They have Effective skills in writing, reading and oral communication (listening and speaking skills) with required clarity. Basic mathematical skills in collecting, communicating and presenting materials based on sound social political and natural environment. Effective team work and safety of self and colleagues. Responsible for self-learning goals.

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. [MES/N2528: Analyse 3D Technology for printing](#)
2. [MES/N0533: Prepare 3D design, digital models and prototypes](#)
3. [MES/N0534: Data preparation and printing](#)
4. [MES/N0535: Operate 3D scanning and printing machinery](#)
5. [MES/N0536: Conduct servicing and repairing equipment](#)
6. [DGT/VSQ/N0102: Employability Skills \(60 Hours\)](#)
7. [MES/N0104: Maintain Workplace Health & Safety](#)

Qualification Pack (QP) Parameters

Sector	Media & Entertainment
Sub-Sector	Print
Occupation	Art and Design

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Country	India
NSQF Level	4
Credits	19
Aligned to NCO/ISCO/ISIC Code	NCO-2015/2141.2400
Minimum Educational Qualification & Experience	11th Class (pass) OR 10th Class (pass plus 1 year NTC/NAC) OR 10th Class (pass and pursuing continuous schooling)
Minimum Level of Education for Training in School	10th Class
Pre-Requisite License or Training	NA
Minimum Job Entry Age	18 Years
Last Reviewed On	NA
Next Review Date	30/03/2027
NSQC Approval Date	31/03/2022
Version	2.0
Reference code on NQR	2022/ME/MESC/05697
NQR Version	2.0

Remarks:

The QP has been rationalised as per NCVET guidelines.

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MES/N2528: Analyse 3D Technology for printing

Description

This unit is about analyzing 3D printing technology and creating a 3D printed object

Scope

The scope covers the following :

- Discuss and demonstrate ways to analyze 3D printing technology
- Create a 3D printed object

Elements and Performance Criteria

Analyze 3D printing technology

To be competent, the user/individual on the job must be able to:

- PC1.** • Analyse 3D print technology- 3D printing or additive manufacturing is a process of making three dimensional
• solid objects from a digital file
- PC2.** creation of a 3D printed object is achieved using additive processes
- PC3.** Identify and install 3D software to be used

Create a 3D printed object

To be competent, the user/individual on the job must be able to:

- PC4.** Slicing - slicing up a 3D model into hundreds or thousands of layers
- PC5.** Demonstration of different stages of AM (Additive Manufacturing) Process
- PC6.** Programming of AM Parts.
- PC7.** Working with 3D printer
- PC8.** Select and use correct CAD formats to manufacture a 3D printed part.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** the Quantity and the quality as per the specification.
- KU2.** to make less wastage.
- KU3.** to increase the production and efficiency of the machine.
- KU4.** to discuss and distribute the work among the team.
- KU5.** to maintain quality control as production scales
- KU6.** Install 3D software
- KU7.** how to select appropriate CAD formats
- KU8.** how to do slicing of a 3D model
- KU9.** different stages of Additive manufacturing process
- KU10.** various AM technologies

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- KU11.** ways to select Laser Sintering Polyjet, Materials for Additive Manufacturing & 3D Printing
- KU12.** various technologies including Computer-Aided Design Technology - Other Associated Technologies, Lasers, Printing Technologies, Programmable Logic Controllers
- KU13.** process of Material Extrusion, Power bed fusion and Sheet Lamination
- KU14.** various slicing tools, Finite Element Analysis
- KU15.** Preparing STLs for 3D Printing
- KU16.** how to process Simulations Using Finite Element Analysis
- KU17.** use of AM to Support Medical Applications, Surgical and Diagnostic Aids, Prosthetics Development
- KU18.** processes of contact and noncontact 3D scanning
- KU19.** how to create and plot assembly and detail views of simple geometrical solid with Dimension, Tolerance & Annotation in 3D Modelling
- KU20.** different measurement to be performed to check the components for functionality and conformance
- KU21.** use of various instruments such as Vernier Caliper, Vernier height gauge, Micrometer, depth gauge, Bevel protector, sine bar and dial test indicator
- KU22.** how to apply process algorithm
- KU23.** keeping 3D printer well lubricated
- KU24.** how to clear any dust and debris from the extruder feeder wheels
- KU25.** how to maintain 3D printer

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** prepare the production report/ documentation including samples to accompany the job docket.
- GS2.** read and understand the user and technical specifications of jobs.
- GS3.** discuss and understand requirements and specifications from the Print Coordinator and the team.
- GS4.** discuss any problems with the brief that could impact the production process and solicit suggestions for resolving them.
- GS5.** To plan and prioritize work according to the requirements.
- GS6.** Complete the job within a period of time by increasing the efficiency of the machine.
- GS7.** make decision for suitable course of action.
- GS8.** quality standards/final output meet customer requirements and organizational standards.
- GS9.** ensure that the work requires technical assistance to meet the desired outcomes and resolve the same.
- GS10.** solve the problems while printing without stoppage of machine unless & until it is necessary.

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Analyze 3D printing technology</i>	20	50	-	-
PC1. • Analyse 3D print technology- 3D printing or additive manufacturing is a process of making three dimensional • solid objects from a digital file	5	-	-	-
PC2. creation of a 3D printed object is achieved using additive processes	10	-	-	-
PC3. Identify and install 3D software to be used	5	-	-	-
<i>Create a 3D printed object</i>	30	-	-	-
PC4. Slicing - slicing up a 3D model into hundreds or thousands of layers	5	-	-	-
PC5. Demonstration of different stages of AM (Additive Manufacturing) Process	10	-	-	-
PC6. Programming of AM Parts.	5	-	-	-
PC7. Working with 3D printer	5	-	-	-
PC8. Select and use correct CAD formats to manufacture a 3D printed part.	5	-	-	-
NOS Total	50	50	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	MES/N2528
NOS Name	Analyse 3D Technology for printing
Sector	Media & Entertainment
Sub-Sector	Print
Occupation	Asset Creation, Art and Design
NSQF Level	4
Credits	TBD
Version	1.0
Last Reviewed Date	31/03/2022
Next Review Date	30/03/2027
NSQC Clearance Date	31/03/2022

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MES/N0533: Prepare 3D design, digital models and prototypes

Description

This unit is about various AM technologies and to use 3D printers to create digital models and prototypes

Scope

The scope covers the following :

- Define various milestones in AM Development
- Demonstrate ways to use 3D printers to create digital models and prototypes

Elements and Performance Criteria

Elaborate various AM technologies and AM processes

To be competent, the user/individual on the job must be able to:

- PC1.** Selection of Laser Sintering Polyjet, Materials for Additive Manufacturing & 3D Printing
- PC2.** Development of AM Technologies, Computers, Computer-Aided Design Technology - Other Associated Technologies, Lasers, Printing Technologies, Programmable Logic Controllers, Materials, Computer
- PC3.**
 - Use of layers , classification of AM Processes, New AM Classification Schemes, Metal Systems, Hybrid
 - Systems, Milestones in AM Development, AM Around the World, Rapid Prototyping Develops into
 - Direct Digital Manufacturing

Prepare 3D design, digital models and prototypes

To be competent, the user/individual on the job must be able to:

- PC4.** Use 3D printers in their design process to create prototypes
- PC5.**
 - Apply additive manufacturing type
 - • Vat Photopolymerisation – Stereolithography, Digital Light processing, Continuous liquid interface production
 - • Material Jetting
 - • Binder Jetting
 - • Material Extrusion – Fused deposition modelling, Fused, Filament fabrication
 - • Power bed fusion: Multi Jet Fusion, Selective laser sintering,
 - • Direct metal laser sintering
 - • Sheet Lamination
 - • Directive energy deposition

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** the Quantity and the quality as per the specification.
- KU2.** to make less wastage.
- KU3.** to increase the production and efficiency of the machine.
- KU4.** to discuss and distribute the work among the team.

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- KU5.** to maintain quality control as production scales
- KU6.** Install 3D software
- KU7.** how to select appropriate CAD formats
- KU8.** how to do slicing of a 3D model
- KU9.** different stages of Additive manufacturing process
- KU10.** various AM technologies
- KU11.** ways to select Laser Sintering Polyjet, Materials for Additive Manufacturing & 3D Printing
- KU12.** various technologies including Computer-Aided Design Technology - Other Associated Technologies, Lasers, Printing Technologies, Programmable Logic Controllers
- KU13.** process of Material Extrusion, Power bed fusion and Sheet Lamination
- KU14.** various slicing tools, Finite Element Analysis
- KU15.** Preparing STLs for 3D Printing
- KU16.** how to process Simulations Using Finite Element Analysis
- KU17.** use of AM to Support Medical Applications, Surgical and Diagnostic Aids, Prosthetics Development
- KU18.** processes of contact and noncontact 3D scanning
- KU19.** how to create and plot assembly and detail views of simple geometrical solid with Dimension, Tolerance & Annotation in 3D Modelling
- KU20.** different measurement to be performed to check the components for functionality and conformance
- KU21.** use of various instruments such as Vernier Caliper, Vernier height gauge, Micrometer, depth gauge, Bevel protector, sine bar and dial test indicator
- KU22.** how to apply process algorithm
- KU23.** keeping 3D printer well lubricated
- KU24.** how to clear any dust and debris from the extruder feeder wheels
- KU25.** how to maintain 3D printer

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** prepare the production report/ documentation including samples to accompany the job docket.
- GS2.** read and understand the user and technical specifications of jobs.
- GS3.** discuss and understand requirements and specifications from the Print Coordinator and the team.
- GS4.** discuss any problems with the brief that could impact the production process and solicit suggestions for resolving them.
- GS5.** To plan and prioritize work according to the requirements.
- GS6.** Complete the job within a period of time by increasing the efficiency of the machine.
- GS7.** make decision for suitable course of action.
- GS8.** quality standards/final output meet customer requirements and organizational standards.

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- GS9.** ensure that the work requires technical assistance to meet the desired outcomes and resolve the same.
- GS10.** solve the problems while printing without stoppage of machine unless & until it is necessary.

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Elaborate various AM technologies and AM processes</i>	15	70	-	-
PC1. Selection of Laser Sintering Polyjet, Materials for Additive Manufacturing & 3D Printing	5	-	-	-
PC2. Development of AM Technologies, Computers, Computer-Aided Design Technology - Other Associated Technologies, Lasers, Printing Technologies, Programmable Logic Controllers, Materials, Computer	5	-	-	-
PC3. • Use of layers , classification of AM Processes, New AM Classification Schemes, Metal Systems, Hybrid • Systems, Milestones in AM Development, AM Around the World, Rapid Prototyping Develops into • Direct Digital Manufacturing	5	-	-	-
<i>Prepare 3D design, digital models and prototypes</i>	15	-	-	-
PC4. Use 3D printers in their design process to create prototypes	5	-	-	-
PC5. • Apply additive manufacturing type • • Vat Photopolymerisation – Stereolithography, Digital Light processing, Continuous liquid interface production • • Material Jetting • • Binder Jetting • • Material Extrusion – Fused deposition modelling, Fused, Filament fabrication • • Power bed fusion: Multi Jet Fusion, Selective laser sintering, • • Direct metal laser sintering • • Sheet Lamination • • Directive energy deposition	10	-	-	-
NOS Total	30	70	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	MES/N0533
NOS Name	Prepare 3D design, digital models and prototypes
Sector	Media & Entertainment
Sub-Sector	Print
Occupation	Art and Design
NSQF Level	4
Credits	2
Version	1.0
Last Reviewed Date	31/03/2022
Next Review Date	30/03/2027
NSQC Clearance Date	31/03/2022

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MES/N0534: Data preparation and printing

Description

This unit is about preparing relevant data and carrying out the process of printing

Scope

The scope covers the following :

- Carry out the process of printing
- Prepare relevant data required for the process of printing

Elements and Performance Criteria

Carry out the process of printing

To be competent, the user/individual on the job must be able to:

- PC1.** Use of slicing tools
- PC2.** Prepare STLs for 3D Printing
- PC3.** Prepare CAD Models with STL file
- PC4.** Process Simulations Using Finite Element Analysis
- PC5.** Functional Testing, Rapid Tooling
- PC6.** Use of AM to Support Medical Applications, Surgical and Diagnostic Aids, Prosthetics Development

Prepare relevant data required for the process of printing

To be competent, the user/individual on the job must be able to:

- PC7.** Outline on Manufacturing, Tissue Engineering and Organ Printing
- PC8.** Able to give Software Support, Product Prototyping, Art, Jewellery
- PC9.** identify opportunities to apply 3D printing Technology for time and cost saving.
- PC10.** Ensure 3D printing task successfully and confirm as deliverable. Also identify Entrepreneurial opportunities in 3D Printing

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** the Quantity and the quality as per the specification.
- KU2.** to make less wastage.
- KU3.** to increase the production and efficiency of the machine.
- KU4.** to discuss and distribute the work among the team.
- KU5.** to maintain quality control as production scales
- KU6.** Install 3D software
- KU7.** how to select appropriate CAD formats

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- KU8.** how to do slicing of a 3D model
- KU9.** different stages of Additive manufacturing process
- KU10.** various AM technologies
- KU11.** ways to select Laser Sintering Polyjet, Materials for Additive Manufacturing & 3D Printing
- KU12.** various technologies including Computer-Aided Design Technology - Other Associated Technologies, Lasers, Printing Technologies, Programmable Logic Controllers
- KU13.** process of Material Extrusion, Powder bed fusion and Sheet Lamination
- KU14.** various slicing tools, Finite Element Analysis
- KU15.** Preparing STLs for 3D Printing
- KU16.** how to process Simulations Using Finite Element Analysis
- KU17.** use of AM to Support Medical Applications, Surgical and Diagnostic Aids, Prosthetics Development
- KU18.** processes of contact and noncontact 3D scanning
- KU19.** how to create and plot assembly and detail views of simple geometrical solid with Dimension, Tolerance & Annotation in 3D Modelling
- KU20.** different measurement to be performed to check the components for functionality and conformance
- KU21.** use of various instruments such as Vernier Caliper, Vernier height gauge, Micrometer, depth gauge, Bevel protector, sine bar and dial test indicator
- KU22.** how to apply process algorithm
- KU23.** keeping 3D printer well lubricated
- KU24.** how to clear any dust and debris from the extruder feeder wheels
- KU25.** how to maintain 3D printer

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** prepare the production report/ documentation including samples to accompany the job docket.
- GS2.** read and understand the user and technical specifications of jobs.
- GS3.** discuss and understand requirements and specifications from the Print Coordinator and the team.
- GS4.** discuss any problems with the brief that could impact the production process and solicit suggestions for resolving them.
- GS5.** To plan and prioritize work according to the requirements.
- GS6.** Complete the job within a period of time by increasing the efficiency of the machine.
- GS7.** make decision for suitable course of action.
- GS8.** quality standards/final output meet customer requirements and organizational standards.
- GS9.** ensure that the work requires technical assistance to meet the desired outcomes and resolve the same.
- GS10.** solve the problems while printing without stoppage of machine unless & until it is necessary.

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Carry out the process of printing</i>	30	50	-	-
PC1. Use of slicing tools	5	-	-	-
PC2. Prepare STLs for 3D Printing	5	-	-	-
PC3. Prepare CAD Models with STL file	5	-	-	-
PC4. Process Simulations Using Finite Element Analysis	5	-	-	-
PC5. Functional Testing, Rapid Tooling	5	-	-	-
PC6. Use of AM to Support Medical Applications, Surgical and Diagnostic Aids, Prosthetics Development	5	-	-	-
<i>Prepare relevant data required for the process of printing</i>	20	-	-	-
PC7. Outline on Manufacturing, Tissue Engineering and Organ Printing	5	-	-	-
PC8. Able to give Software Support, Product Prototyping, Art, Jewellery	5	-	-	-
PC9. identify opportunities to apply 3D printing Technology for time and cost saving.	5	-	-	-
PC10. Ensure 3D printing task successfully and confirm as deliverable. Also identify Entrepreneurial opportunities in 3D Printing	5	-	-	-
NOS Total	50	50	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	MES/N0534
NOS Name	Data preparation and printing
Sector	Media & Entertainment
Sub-Sector	Print
Occupation	Art and Design
NSQF Level	4
Credits	3
Version	1.0
Last Reviewed Date	31/03/2022
Next Review Date	30/03/2027
NSQC Clearance Date	31/03/2022

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MES/N0535: Operate 3D scanning and printing machinery

Description

This unit covers how to create and plot assembly and detail views of simple geometrical solid with Dimension, Tolerance & Annotation in 3D Modelling

Scope

The scope covers the following :

- Demonstrate ways to operate 3D scanning
- Develop a prototype/end use product

Elements and Performance Criteria

Operate 3D scanning

To be competent, the user/individual on the job must be able to:

- PC1.** • Scan the content Contact 3D scanning employs, some kind of arm, like a robotic arm, equipped with a probe.
• Non-contact 3D scanning involves collecting radiation originating from the target object and can employ active or passive techniques
- PC2.** Create and plot assembly and detail views of simple geometrical solid with Dimension, Tolerance & Annotation in 3D Modelling
- PC3.** Perform different measurements with desired accuracy to check the components for functionality and conformance to defined standard using different instruments. [Different measurement: linear, taper, surface roughness, angular, thread; Different instruments: Vernier Caliper, Vernier height gauge, Micrometer, depth gauge, Bevel protector, sine bar, dial test indicator]
- PC4.** Explain Additive Manufacturing (AM) Technology and emerging trends in Additive Manufacturing

Create a prototype/end use product

To be competent, the user/individual on the job must be able to:

- PC5.** Develop a prototype/ end use product
- PC6.** Apply process algorithm (Slicing Software)
- PC7.** Make a simple fixture for functional requirement.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** the Quantity and the quality as per the specification.
- KU2.** to make less wastage.
- KU3.** to increase the production and efficiency of the machine.
- KU4.** to discuss and distribute the work among the team.

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- KU5.** to maintain quality control as production scales
- KU6.** Install 3D software
- KU7.** how to select appropriate CAD formats
- KU8.** how to do slicing of a 3D model
- KU9.** different stages of Additive manufacturing process
- KU10.** various AM technologies
- KU11.** ways to select Laser Sintering Polyjet, Materials for Additive Manufacturing & 3D Printing
- KU12.** various technologies including Computer-Aided Design Technology - Other Associated Technologies, Lasers, Printing Technologies, Programmable Logic Controllers
- KU13.** process of Material Extrusion, Power bed fusion and Sheet Lamination
- KU14.** various slicing tools, Finite Element Analysis
- KU15.** Preparing STLs for 3D Printing
- KU16.** how to process Simulations Using Finite Element Analysis
- KU17.** use of AM to Support Medical Applications, Surgical and Diagnostic Aids, Prosthetics Development
- KU18.** processes of contact and noncontact 3D scanning
- KU19.** how to create and plot assembly and detail views of simple geometrical solid with Dimension, Tolerance & Annotation in 3D Modelling
- KU20.** different measurement to be performed to check the components for functionality and conformance
- KU21.** use of various instruments such as Vernier Caliper, Vernier height gauge, Micrometer, depth gauge, Bevel protector, sine bar and dial test indicator
- KU22.** how to apply process algorithm
- KU23.** keeping 3D printer well lubricated
- KU24.** how to clear any dust and debris from the extruder feeder wheels
- KU25.** how to maintain 3D printer

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** prepare the production report/ documentation including samples to accompany the job docket.
- GS2.** read and understand the user and technical specifications of jobs.
- GS3.** discuss and understand requirements and specifications from the Print Coordinator and the team.
- GS4.** discuss any problems with the brief that could impact the production process and solicit suggestions for resolving them.
- GS5.** To plan and prioritize work according to the requirements.
- GS6.** Complete the job within a period of time by increasing the efficiency of the machine.
- GS7.** make decision for suitable course of action.
- GS8.** quality standards/final output meet customer requirements and organizational standards.

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- GS9.** ensure that the work requires technical assistance to meet the desired outcomes and resolve the same.
- GS10.** solve the problems while printing without stoppage of machine unless & until it is necessary.

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Operate 3D scanning</i>	20	65	-	-
PC1. - Scan the content Contact 3D scanning employs, some kind of arm, like a robotic arm, equipped with a probe. - Non-contact 3D scanning involves collecting radiation originating from the target object and can employ active or passive techniques	5	-	-	-
PC2. Create and plot assembly and detail views of simple geometrical solid with Dimension, Tolerance & Annotation in 3D Modelling	5	-	-	-
PC3. Perform different measurements with desired accuracy to check the components for functionality and conformance to defined standard using different instruments. [Different measurement: linear, taper, surface roughness, angular, thread; Different instruments: Vernier Caliper, Vernier height gauge, Micrometer, depth gauge, Bevel protector, sine bar, dial test indicator]	5	-	-	-
PC4. Explain Additive Manufacturing (AM) Technology and emerging trends in Additive Manufacturing	5	-	-	-
<i>Create a prototype/end use product</i>	15	-	-	-
PC5. Develop a prototype/ end use product	5	-	-	-
PC6. Apply process algorithm (Slicing Software)	5	-	-	-
PC7. Make a simple fixture for functional requirement.	5	-	-	-
NOS Total	35	65	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	MES/N0535
NOS Name	Operate 3D scanning and printing machinery
Sector	Media & Entertainment
Sub-Sector	Print
Occupation	Art and Design
NSQF Level	4
Credits	3
Version	1.0
Last Reviewed Date	31/03/2022
Next Review Date	30/03/2027
NSQF Clearance Date	31/03/2022

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MES/N0536: Conduct servicing and repairing equipment

Description

This unit covers service and repairing of various components of 3D printer and other equipment.

Scope

The scope covers the following :

- Demonstrate ways to conduct service and repairing of various components of 3D printer and other equipment.

Elements and Performance Criteria

Servicing and repairing equipment

To be competent, the user/individual on the job must be able to:

- PC1.** Keep 3D printer well lubricated
- PC2.** Dust the printer and its components regularly: As the 3D printer moves around, the seals on the bearings attached to each carriage will sweep dust to the limits of the motion system. Its fans actually collect dust and can build up a sort of cobweb on them and anything near them including around the hot end.
- PC3.** Check for loose nuts and bolts
- PC4.** Clear any dust and debris from the extruder feeder wheels
- PC5.** Tidy up and remove loose bits of 3D printing debris
- PC6.** Check for overheated and deformed 3D printed parts
- PC7.** Tighten up belts, Maintain and replace your bowden tube
- PC8.** Clean or replace your nozzle often

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** the Quantity and the quality as per the specification.
- KU2.** to make less wastage.
- KU3.** to increase the production and efficiency of the machine.
- KU4.** to discuss and distribute the work among the team.
- KU5.** to maintain quality control as production scales
- KU6.** Install 3D software
- KU7.** how to select appropriate CAD formats
- KU8.** how to do slicing of a 3D model
- KU9.** different stages of Additive manufacturing process
- KU10.** various AM technologies
- KU11.** ways to select Laser Sintering Polyjet, Materials for Additive Manufacturing & 3D Printing

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- KU12.** various technologies including Computer-Aided Design Technology - Other Associated Technologies, Lasers, Printing Technologies, Programmable Logic Controllers
- KU13.** process of Material Extrusion, Power bed fusion and Sheet Lamination
- KU14.** various slicing tools, Finite Element Analysis
- KU15.** Preparing STLs for 3D Printing
- KU16.** how to process Simulations Using Finite Element Analysis
- KU17.** use of AM to Support Medical Applications, Surgical and Diagnostic Aids, Prosthetics Development
- KU18.** processes of contact and noncontact 3D scanning
- KU19.** how to create and plot assembly and detail views of simple geometrical solid with Dimension, Tolerance & Annotation in 3D Modelling
- KU20.** different measurement to be performed to check the components for functionality and conformance
- KU21.** use of various instruments such as Vernier Caliper, Vernier height gauge, Micrometer, depth gauge, Bevel protector, sine bar and dial test indicator
- KU22.** how to apply process algorithm
- KU23.** keeping 3D printer well lubricated
- KU24.** how to clear any dust and debris from the extruder feeder wheels
- KU25.** how to maintain 3D printer

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** prepare the production report/ documentation including samples to accompany the job docket.
- GS2.** read and understand the user and technical specifications of jobs.
- GS3.** discuss and understand requirements and specifications from the Print Coordinator and the team.
- GS4.** discuss any problems with the brief that could impact the production process and solicit suggestions for resolving them.
- GS5.** To plan and prioritize work according to the requirements.
- GS6.** Complete the job within a period of time by increasing the efficiency of the machine.
- GS7.** make decision for suitable course of action.
- GS8.** quality standards/final output meet customer requirements and organizational standards.
- GS9.** ensure that the work requires technical assistance to meet the desired outcomes and resolve the same.
- GS10.** solve the problems while printing without stoppage of machine unless & until it is necessary.

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Servicing and repairing equipment</i>	40	60	-	-
PC1. Keep 3D printer well lubricated	5	-	-	-
PC2. Dust the printer and its components regularly: As the 3D printer moves around, the seals on the bearings attached to each carriage will sweep dust to the limits of the motion system. Its fans actually collect dust and can build up a sort of cobweb on them and anything near them including around the hot end.	5	-	-	-
PC3. Check for loose nuts and bolts	5	-	-	-
PC4. Clear any dust and debris from the extruder feeder wheels	5	-	-	-
PC5. Tidy up and remove loose bits of 3D printing debris	5	-	-	-
PC6. Check for overheated and deformed 3D printed parts	5	-	-	-
PC7. Tighten up belts, Maintain and replace your bowden tube	5	-	-	-
PC8. Clean or replace your nozzle often	5	-	-	-
NOS Total	40	60	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	MES/N0536
NOS Name	Conduct servicing and repairing equipment
Sector	Media & Entertainment
Sub-Sector	Print
Occupation	Art and Design
NSQF Level	4
Credits	2
Version	1.0
Last Reviewed Date	31/03/2022
Next Review Date	30/03/2027
NSQC Clearance Date	31/03/2022

Qualification Pack

DGT/VSQ/N0102: Employability Skills (60 Hours)

Description

This unit is about employability skills, Constitutional values, becoming a professional in the 21st Century, digital, financial, and legal literacy, diversity and Inclusion, English and communication skills, customer service, entrepreneurship, and apprenticeship, getting ready for jobs and career development.

Scope

The scope covers the following :

- Introduction to Employability Skills
- Constitutional values - Citizenship
- Becoming a Professional in the 21st Century
- Basic English Skills
- Career Development & Goal Setting
- Communication Skills
- Diversity & Inclusion
- Financial and Legal Literacy
- Essential Digital Skills
- Entrepreneurship
- Customer Service
- Getting ready for Apprenticeship & Jobs

Elements and Performance Criteria

Introduction to Employability Skills

To be competent, the user/individual on the job must be able to:

- PC1.** identify employability skills required for jobs in various industries
- PC2.** identify and explore learning and employability portals

Constitutional values – Citizenship

To be competent, the user/individual on the job must be able to:

- PC3.** recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.
- PC4.** follow environmentally sustainable practices

Becoming a Professional in the 21st Century

To be competent, the user/individual on the job must be able to:

- PC5.** recognize the significance of 21st Century Skills for employment
- PC6.** practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life

Basic English Skills

To be competent, the user/individual on the job must be able to:

Qualification Pack

- PC7.** use basic English for everyday conversation in different contexts, in person and over the telephone
- PC8.** read and understand routine information, notes, instructions, mails, letters etc. written in English
- PC9.** write short messages, notes, letters, e-mails etc. in English

Career Development & Goal Setting

To be competent, the user/individual on the job must be able to:

- PC10.** understand the difference between job and career
- PC11.** prepare a career development plan with short- and long-term goals, based on aptitude

Communication Skills

To be competent, the user/individual on the job must be able to:

- PC12.** follow verbal and non-verbal communication etiquette and active listening techniques in various settings
- PC13.** work collaboratively with others in a team

Diversity & Inclusion

To be competent, the user/individual on the job must be able to:

- PC14.** communicate and behave appropriately with all genders and PwD
- PC15.** escalate any issues related to sexual harassment at workplace according to POSH Act

Financial and Legal Literacy

To be competent, the user/individual on the job must be able to:

- PC16.** select financial institutions, products and services as per requirement
- PC17.** carry out offline and online financial transactions, safely and securely
- PC18.** identify common components of salary and compute income, expenses, taxes, investments etc
- PC19.** identify relevant rights and laws and use legal aids to fight against legal exploitation

Essential Digital Skills

To be competent, the user/individual on the job must be able to:

- PC20.** operate digital devices and carry out basic internet operations securely and safely
- PC21.** use e-mail and social media platforms and virtual collaboration tools to work effectively
- PC22.** use basic features of word processor, spreadsheets, and presentations

Entrepreneurship

To be competent, the user/individual on the job must be able to:

- PC23.** identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research
- PC24.** develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion
- PC25.** identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity

Customer Service

To be competent, the user/individual on the job must be able to:

- PC26.** identify different types of customers
- PC27.** identify and respond to customer requests and needs in a professional manner.

Qualification Pack

PC28. follow appropriate hygiene and grooming standards

Getting ready for apprenticeship & Jobs

To be competent, the user/individual on the job must be able to:

PC29. create a professional Curriculum vitae (Résumé)

PC30. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively

PC31. apply to identified job openings using offline /online methods as per requirement

PC32. answer questions politely, with clarity and confidence, during recruitment and selection

PC33. identify apprenticeship opportunities and register for it as per guidelines and requirements

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. need for employability skills and different learning and employability related portals

KU2. various constitutional and personal values

KU3. different environmentally sustainable practices and their importance

KU4. Twenty first (21st) century skills and their importance

KU5. how to use English language for effective verbal (face to face and telephonic) and written communication in formal and informal set up

KU6. importance of career development and setting long- and short-term goals

KU7. about effective communication

KU8. POSH Act

KU9. Gender sensitivity and inclusivity

KU10. different types of financial institutes, products, and services

KU11. how to compute income and expenditure

KU12. importance of maintaining safety and security in offline and online financial transactions

KU13. different legal rights and laws

KU14. different types of digital devices and the procedure to operate them safely and securely

KU15. how to create and operate an e- mail account and use applications such as word processors, spreadsheets etc.

KU16. how to identify business opportunities

KU17. types and needs of customers

KU18. how to apply for a job and prepare for an interview

KU19. apprenticeship scheme and the process of registering on apprenticeship portal

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. read and write different types of documents/instructions/correspondence

GS2. communicate effectively using appropriate language in formal and informal settings

Qualification Pack

- GS3.** behave politely and appropriately with all
- GS4.** how to work in a virtual mode
- GS5.** perform calculations efficiently
- GS6.** solve problems effectively
- GS7.** pay attention to details
- GS8.** manage time efficiently
- GS9.** maintain hygiene and sanitization to avoid infection

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Introduction to Employability Skills</i>	1	1	-	-
PC1. identify employability skills required for jobs in various industries	-	-	-	-
PC2. identify and explore learning and employability portals	-	-	-	-
<i>Constitutional values – Citizenship</i>	1	1	-	-
PC3. recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.	-	-	-	-
PC4. follow environmentally sustainable practices	-	-	-	-
<i>Becoming a Professional in the 21st Century</i>	2	4	-	-
PC5. recognize the significance of 21st Century Skills for employment	-	-	-	-
PC6. practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life	-	-	-	-
<i>Basic English Skills</i>	2	3	-	-
PC7. use basic English for everyday conversation in different contexts, in person and over the telephone	-	-	-	-
PC8. read and understand routine information, notes, instructions, mails, letters etc. written in English	-	-	-	-
PC9. write short messages, notes, letters, e-mails etc. in English	-	-	-	-
<i>Career Development & Goal Setting</i>	1	2	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. understand the difference between job and career	-	-	-	-
PC11. prepare a career development plan with short- and long-term goals, based on aptitude	-	-	-	-
<i>Communication Skills</i>	2	2	-	-
PC12. follow verbal and non-verbal communication etiquette and active listening techniques in various settings	-	-	-	-
PC13. work collaboratively with others in a team	-	-	-	-
<i>Diversity & Inclusion</i>	1	2	-	-
PC14. communicate and behave appropriately with all genders and PwD	-	-	-	-
PC15. escalate any issues related to sexual harassment at workplace according to POSH Act	-	-	-	-
<i>Financial and Legal Literacy</i>	2	3	-	-
PC16. select financial institutions, products and services as per requirement	-	-	-	-
PC17. carry out offline and online financial transactions, safely and securely	-	-	-	-
PC18. identify common components of salary and compute income, expenses, taxes, investments etc	-	-	-	-
PC19. identify relevant rights and laws and use legal aids to fight against legal exploitation	-	-	-	-
<i>Essential Digital Skills</i>	3	4	-	-
PC20. operate digital devices and carry out basic internet operations securely and safely	-	-	-	-
PC21. use e- mail and social media platforms and virtual collaboration tools to work effectively	-	-	-	-
PC22. use basic features of word processor, spreadsheets, and presentations	-	-	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Entrepreneurship</i>	2	3	-	-
PC23. identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research	-	-	-	-
PC24. develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion	-	-	-	-
PC25. identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity	-	-	-	-
<i>Customer Service</i>	1	2	-	-
PC26. identify different types of customers	-	-	-	-
PC27. identify and respond to customer requests and needs in a professional manner.	-	-	-	-
PC28. follow appropriate hygiene and grooming standards	-	-	-	-
<i>Getting ready for apprenticeship & Jobs</i>	2	3	-	-
PC29. create a professional Curriculum vitae (Résumé)	-	-	-	-
PC30. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively	-	-	-	-
PC31. apply to identified job openings using offline /online methods as per requirement	-	-	-	-
PC32. answer questions politely, with clarity and confidence, during recruitment and selection	-	-	-	-
PC33. identify apprenticeship opportunities and register for it as per guidelines and requirements	-	-	-	-
NOS Total	20	30	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	DGT/VSQ/N0102
NOS Name	Employability Skills (60 Hours)
Sector	Cross Sectoral
Sub-Sector	Professional Skills
Occupation	Employability
NSQF Level	4
Credits	2
Version	1.0
Last Reviewed Date	12/03/2026
Next Review Date	12/03/2029
NSQF Clearance Date	12/03/2026

Qualification Pack

MES/N0104: Maintain Workplace Health & Safety

Description

This OS unit is about contributing towards maintaining a healthy, safe and secure working environment

Elements and Performance Criteria

Understanding the health, safety and security risks prevalent in the workplace

To be competent, the user/individual on the job must be able to:

- PC1.** understand and comply with the organizations current health, safety and security policies and procedures
- PC2.** understand the safe working practices pertaining to own occupation
- PC3.** understand the government norms and policies relating to health and safety including emergency procedures for illness, accidents, fires or others which may involve evacuation of the premises
- PC4.** participate in organization health and safety knowledge sessions and drills

Knowing the people responsible for health and safety and the resources available

To be competent, the user/individual on the job must be able to:

- PC5.** identify the people responsible for health and safety in the workplace, including those to contact in case of an emergency
- PC6.** identify security signals e.g. fire alarms and places such as staircases, fire warden stations, first aid and medical rooms

Identifying and reporting risks

To be competent, the user/individual on the job must be able to:

- PC7.** identify aspects of your workplace that could cause potential risk to own and others health and safety
- PC8.** ensure own personal health and safety, and that of others in the workplace through precautionary measures
- PC9.** identify and recommend opportunities for improving health, safety, and security to the designated person
- PC10.** report any hazards outside the individuals authority to the relevant person in line with organizational procedures and warn other people who may be affected

Complying with procedures in the event of an emergency

To be competent, the user/individual on the job must be able to:

- PC11.** follow organizations emergency procedures for accidents, fires or any other natural calamity in case of a hazard
- PC12.** identify and correct risks like illness, accidents, fires or any other natural calamity safely and within the limits of individuals authority

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

Qualification Pack

- KU1.** Organizations norms and policies relating to health and safety
- KU2.** Government norms and policies regarding health and safety and related emergency procedures
- KU3.** Limits of authority while dealing with risks/ hazards
- KU4.** The importance of maintaining high standards of health and safety at a workplace
- KU5.** The different types of health and safety hazards in a workplace
- KU6.** Safe working practices for own job role
- KU7.** Evacuation procedures and other arrangements for handling risks
- KU8.** Names and contact numbers of people responsible for health and safety in a workplace
- KU9.** How to summon medical assistance and the emergency services, where necessary
- KU10.** Vendors or manufacturers instructions for maintaining health and safety while using equipment, systems and/or machines

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** how to write and provide feedback regarding health and safety to the concerned people
- GS2.** how to write and highlight potential risks or report a hazard to the concerned people
- GS3.** read instructions, policies, procedures and norms relating to health and safety
- GS4.** highlight potential risks and report hazards to the designated people
- GS5.** listen and communicate information with all anyone concerned or affected
- GS6.** make decisions on a suitable course of action or plan
- GS7.** plan and organize people and resources to deal with risks/ hazards that lie within the scope of ones individual authority
- GS8.** apply problem solving approaches in different situations
- GS9.** understand hazards that fall within the scope of individual authority and report all hazards that may supersede ones authority
- GS10.** apply balanced judgments in different situations
- GS11.** How to write and provide feedback regarding health and safety to the concerned people
- GS12.** How to write and highlight potential risks or report a hazard to the concerned people
- GS13.** Read instructions, policies, procedures and norms relating to health and safety
- GS14.** Highlight potential risks and report hazards to the designated people
- GS15.** Listen and communicate information with all anyone concerned or affected
- GS16.** Make decisions on a suitable course of action or plan
- GS17.** Plan and organize people and resources to deal with risks/ hazards that lie within the scope of ones individual authority
- GS18.** Apply problem solving approaches in different situations
- GS19.** build and maintain positive and effective relationships with colleges and customers
- GS20.** analyze data and activities
- GS21.** Understand hazards that fall within the scope of individual authority and report all hazards that may supersede ones authority



Qualification Pack

GS22. Apply balanced judgments in different situations

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Understanding the health, safety and security risks prevalent in the workplace</i>	15	15	-	-
PC1. understand and comply with the organizations current health, safety and security policies and procedures	5	5	-	-
PC2. understand the safe working practices pertaining to own occupation	5	5	-	-
PC3. understand the government norms and policies relating to health and safety including emergency procedures for illness, accidents, fires or others which may involve evacuation of the premises	3	2	-	-
PC4. participate in organization health and safety knowledge sessions and drills	2	3	-	-
<i>Knowing the people responsible for health and safety and the resources available</i>	10	10	-	-
PC5. identify the people responsible for health and safety in the workplace, including those to contact in case of an emergency	5	5	-	-
PC6. identify security signals e.g. fire alarms and places such as staircases, fire warden stations, first aid and medical rooms	5	5	-	-
<i>Identifying and reporting risks</i>	18	17	-	-
PC7. identify aspects of your workplace that could cause potential risk to own and others health and safety	5	5	-	-
PC8. ensure own personal health and safety, and that of others in the workplace through precautionary measures	5	5	-	-
PC9. identify and recommend opportunities for improving health, safety, and security to the designated person	3	2	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. report any hazards outside the individuals authority to the relevant person in line with organizational procedures and warn other people who may be affected	5	5	-	-
<i>Complying with procedures in the event of an emergency</i>	7	8	-	-
PC11. follow organizations emergency procedures for accidents, fires or any other natural calamity in case of a hazard	5	5	-	-
PC12. identify and correct risks like illness, accidents, fires or any other natural calamity safely and within the limits of individuals authority	2	3	-	-
NOS Total	50	50	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	MES/N0104
NOS Name	Maintain Workplace Health & Safety
Sector	Media & Entertainment
Sub-Sector	Film, Television, Animation, Gaming, Radio, Advertising
Occupation	Ad sales/Account Management/Scheduling/Traffic
NSQF Level	5
Credits	2
Version	1.0
Last Reviewed Date	10/09/2022
Next Review Date	29/06/2027
NSQC Clearance Date	10/09/2022

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

1. Criteria for assessment for each Qualification Pack will be created by the Sector Skill Council. Each Performance Criteria (PC) will be assigned marks proportional to its importance in NOS. SSC will also lay down proportion of marks for Theory and Skills Practical for each PC.
2. The assessment for the theory part will be based on knowledge bank of questions created by the SSC.
3. Assessment will be conducted for all compulsory NOS, and where applicable, on the selected elective/option NOS/set of NOS.
4. Individual assessment agencies will create unique question papers for theory part for each candidate at each examination/training center (as per assessment criteria below).
5. Individual assessment agencies will create unique evaluations for skill practical for every student at each examination/training center based on this criterion.
6. To pass the Qualification Pack, every trainee should score a minimum of 70% of aggregate marks to successfully clear the assessment.
7. In case of unsuccessful completion, the trainee may seek reassessment on the Qualification Pack.

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Minimum Aggregate Passing % at QP Level : 70

(Please note: Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Assessment Weightage

Compulsory NOS

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
MES/N2528.Analyse 3D Technology for printing	50	50	0	0	100	20
MES/N0533.Prepare 3D design, digital models and prototypes	30	70	-	-	100	20
MES/N0534.Data preparation and printing	50	50	-	-	100	15
MES/N0535.Operate 3D scanning and printing machinery	35	65	-	-	100	10
MES/N0536.Conduct servicing and repairing equipment	40	60	-	-	100	15
DGT/VSQ/N0102.Employability Skills (60 Hours)	20	30	-	-	50	10
MES/N0104.Maintain Workplace Health & Safety	50	50	-	-	100	10
Total	275	375	-	-	650	100

Qualification Pack

Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training
AA	Assessment Agency
AB	Awarding Body
ISCO	International Standard Classification of Occupations
NCO	National Classification of Occupations
NCrF	National Credit Framework

Qualification Pack

Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.

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Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.
Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.
National Occupational Standard	A formal outcome of an assessment and validation process which is obtained when a competent body determinesthat an individual has achieved learning outcomesto given standards
Qualification	A formal outcome of an assessment and validation process which is obtained when a competent body determinesthat an individual has achieved learning outcomesto given standards
Qualification File	AQualification File is a template designed to capture necessary information of a Qualification from the perspective of NSQF compliance. The Qualification File will be normally submitted by the awarding body forthe qualification